

WORKING DRAFT

July 30, 2026

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Todo list



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**Aligning Large Language Models to
publicly collected values**

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Submitted in partial fulfilment of the requirements for
Master of Artificial Intelligence.

Abstract

A short description of the project goes here.

Acknowledgments

Any acknowledgments should go in here, between the title page and the table of contents. The acknowledgments do not form a proper chapter, and so don't get a number or appear in the table of contents.

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Chapter 1

Introduction

This chapter gives an introduction to the project report.
5-6 pages with a full summary

1.1 Report overview

Following on from this introduction chapter there are 5 more chapters to the report. The next chapter ?? gives the key background

This report is structured into 3 main chapters, each focusing on a specific part of the project. For each chapter of the project it explains the methods used and the results obtained. Each chapter is self contained and can be read somewhat independently of the others. The 3 main chapters are:

- ??: Fine-tuning of the LLM on WVS data
- ??: Behavioural analysis of the fine-tuned and base LLMs
- ??: Public consultation on the alignment of the LLMs

The final chapter ?? concludes the report with a summary of the findings, conclusions drawn from the project, key limitations and recommendations for future work.

Chapter 2

Background

A related works section.

Could be removed from word count.

Includes a biref (few pages) summary of the important bits of the related works and how they relate to this project.

Chapter 3

Fine-tuning

This chapter will talk about the fine-tuning process used.

It will mention clustering of WVS data prompts and format of the data finetuning methods finetuning evaluations (survey modelling and response distributions)

The goal of this chapter is to show that an open weight model can be fine-tuned with value survey data to produce a model that responds more like the respondents it is aligned to when answering survey questions.

In section ?? we will find that ...

In section ?? and section ?? I describe the methods used to create the dataset and fine-tune the model.

3.1 Fine-tuning on diverse clusters with appreciable training

3.2 The clusters of survey respondents

To test for inter group differences we need to cluster the survey respondents.

The clusters are distinct

We can assign to the clusters with a high level of confidence

3.2.1 Efficacy of fine-tuning

Discuss the various methods of evaluating the efficacy of fine-tuning, including: - Survey modelling (how well the model predicts survey responses) - Response distributions (how well the model's response distributions match the survey data)

Also explore other concepts of: - Is fine-tuning robust to reasoning on and off? - Does system prompt matter?

3.3 Creating the training dataset

We choose the WVS dataset (free, open and sufficient).

3.3.1 Clustering of survey data

We did clustering using LCA method

This is what the clustering looked like...

3.3.2 Structure of the training data

System prompts and user prompts, how they were selected.

3.4 Fine-tuning methods

To better understand the effects on downstream behaviour 2 types of fine-tuning were done. First is modal fine-tuning where the goal is to have a model that responds in a way that is like the most common response in the survey data for that sub population. The second is distributional fine-tuning where the goal is to have a model that responds in a way that is like the distribution of responses in the survey data for that sub population.

Given the use of LLM reasoning in downstream applications allows for more uncertainty (citation needed, i.e the intuition that the reasoning gives it more chance of randomness and therefore can better model uncertainty.

3.4.1 Modal fine-tuning

To do modal fine-tuning the model is trained used SFT with the expected text response being the most common response in the survey data for that sub population.

A description of SFT

3.4.2 Distributional fine-tuning

To do distributional fine-tuning two methods were used. Firstly was SFT with the expected text response being a random sample from the distribution of responses in the survey data for that sub population. Secondly was using a loss function that is based on the difference between the model's response distribution and the survey data response distribution. This loss function is based on cross-entropy

Description of the fine-tuning process and the loss function used.

Chapter 4

Simulation

A section discussion the behavioural simulations of the model

The selection of scenarios The harness used How summaries of the results were generated
A method for mapping to value dimensions The results of the simulations (compared to clusters are there behavioural clusters)

Chapter 5

Survey

A discussion of the survey

How the questions were selected for assign to cluster

How the scenarios were chosen

Sampling methods

Finding of how well aligned the model is

Chapter 6

Conclusions

The conclusions are presented in this Chapter.

